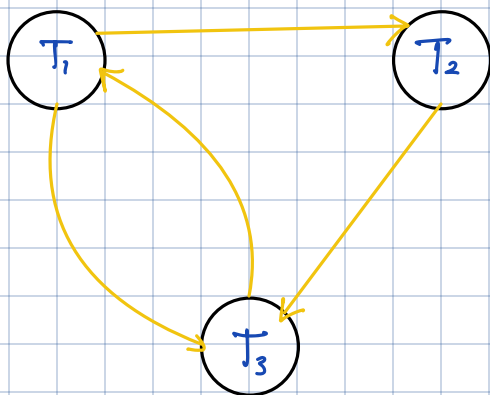


Q22.

(a)

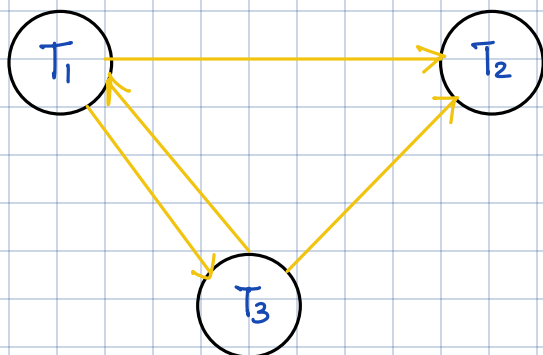
T_1	T_2	T_3
$r_1(x)$		$r_3(x)$
$w_1(x)$	$r_2(x)$	$w_3(x)$



NOT conflict serializable

(b)

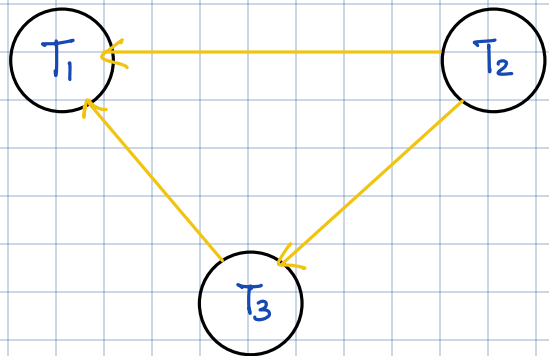
T_1	T_2	T_3
$r_1(x)$		$r_3(x)$
$w_1(x)$	$r_2(x)$	$w_3(x)$



No

(c)

T_1	T_2	T_3
$r_1(x)$ $w_1(x)$	$r_2(x)$	$r_3(x)$ $w_3(x)$

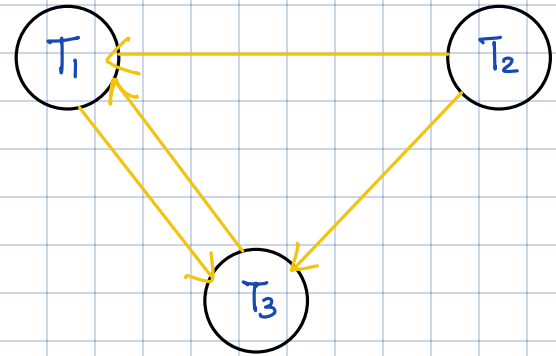


Serializable

Serial Order: $T_2 T_3 T_1$

(d)

T_1	T_2	T_3
$r_1(x)$ $w_1(x)$	$r_2(x)$	$r_3(x)$ $w_3(x)$



14.

For both schedules, we have $M = 2$ and $N = 2$, with an implied consistency rule that X shouldn't exceed 90.

Let me analyze each schedule with the modified T_2 :

Schedule 20.3(a) with modified T_2 :

This is the "lost update problem" schedule where:

- T_1 reads X , calculates $X = X - 2$
- T_2 reads X , calculates $X = X + 2$
- T_2 checks if $X > 90$, then writes X if not
- T_1 's write to X happens (potentially overwriting T_2 's change)
- T_1 updates Y

If we assume X starts with value 80:

- T_1 reads X (80), calculates $X = 78$
- T_2 reads X (80), calculates $X = 82$
- T_2 checks $82 < 90$, so it writes $X = 82$
- T_1 writes $X = 78$ (overwriting T_2 's update)
- T_1 completes with Y

The final value of X is 78, which obeys the constraint ($X < 90$).

Schedule 20.3(b) with modified T_2 :

This is the "temporary update problem" schedule where:

- T_1 reads X, calculates and writes $X = X - 2$
- T_1 fails (requiring rollback)
- T_2 reads X (with the temporary value), calculates $X = X + 2$
- T_2 checks if $X > 90$, then writes X if not

If we assume X starts with value 80:

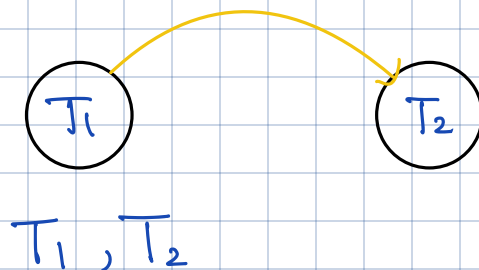
- T_1 reads X (80), calculates and writes $X = 78$
- T_2 reads the temporary X (78), calculates $X = 80$
- T_2 checks $80 < 90$, so it writes $X = 80$
- T_1 fails and must rollback X to 80

The final value of X is 80, which obeys the constraint ($X < 90$).

In both cases, adding the condition doesn't change the fundamental problems, but it does ensure the consistency rule is maintained by preventing writes when X would exceed 90.

Book Example

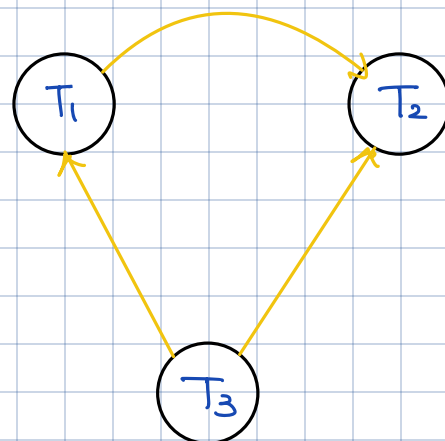
T_1	T_2
read (A) ● $A := A - 50$ write (A) ●	read (A) ● ● $temp := A * 0.1$ $A := A - temp$ write (A) ● ●
read (B) ● $B := B + 50$ write (B) ● commit	read (B) ● $B := B + temp$ write (B) ● commit



① 23.

S_1

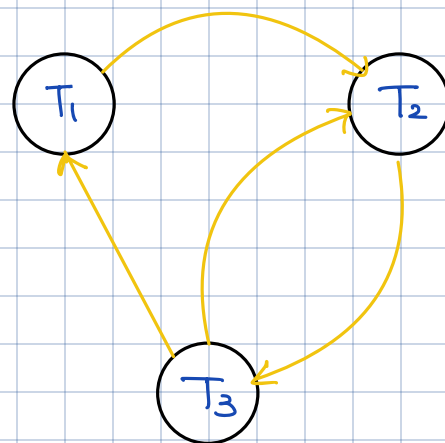
T_1	T_2	T_3
$r_1(X) \cdot$	$r_2(Z) \cdot$	
$r_1(Z) \cdot x$		
$w_1(X) x \checkmark \cdot$		
	$r_2(X) x$	$r_3(X) x x \cdot$
	$w_2(Z) x \checkmark$	$r_3(Y) x x \cdot$
	$w_2(Y) x \checkmark \checkmark$	$w_3(Y) x x \cdot$



T_3, T_1, T_2

S_2

T_1	T_2	T_3
$r_1(X) \cdot$	$r_2(Z) x \cdot$	
$r_1(Z) x x \cdot$	$r_2(Y) x \cdot$	$r_3(X) x x \cdot$
$w_1(X) x \checkmark \cdot$		$r_3(X) x x x \cdot$
	$w_2(Z) x \checkmark \cdot$	$w_3(Y) x x \checkmark \cdot$
	$w_2(Y) x \checkmark x$	



NOT serializable

17.6 Consider the precedence graph of Figure 17.16. Is the corresponding schedule conflict serializable? Explain your answer.

Answer:

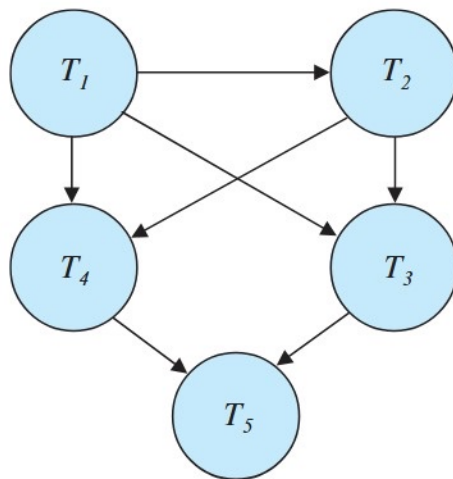


Figure 17.16 Precedence graph for Practice Exercise 17.6.

T_1, T_2, T_3, T_4, T_5

T_1, T_2, T_4, T_3, T_5